communications engineering

A Nature Portfolio journal

Article

https://doi.org/10.1038/s44172-025-00578-0

Human-led truck platooning with lane-changing capability for more efficient logistics: a framework and implementation

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Truck platooning promises to enhance the efficiency of logistics, but commercial operation is hampered by safety and economic concerns. Human-lead truck platooning can mitigate these challenges by leveraging a human driver's expertise. However, existing human-lead truck platooning is limited to longitudinal control and lacks the lane-changing capability, which restricts logistical efficiency. To address this, we build upon previous research to propose a human-lead truck platooning method with lane-changing capability. The platoon leader is controlled by a skilled human driver, who is responsible for leading the following automated trucks. The human-lead platoon is enabled to cruise, lane-change, and obstacle avoidance, leveraging the driver's expertise to mitigate safety risks in long-tail scenarios. Drivers of the following trucks are not needed, reducing labor costs. The proposed method has been implemented in commercial operations at the world's largest port, Shanghai Yangshan Port, achieving an annual transport volume of 200,000 Twenty-foot Equivalent Units. It highlights a route for large-scale truck platooning implementation, potentially reshaping freight-transport operations.

Greenhouse Gas (GHG) emissions are a major driver of climate change, notably contributing to global warming[1–3]. There is a growing consensus across all industrial sectors on the need for decarbonization. The transportation sector, in particular, plays a important role in climate change, consuming more than 100 billion gallons of fossil fuels[4] and accounting for 27% of GHG emissions[5]. Within this sector, road transportation is especially responsible for approximately 72% of fuel consumption and over 80% of CO_2 emissions[6]. To attain carbon neutrality in anthropogenic endeavors, there must be a concentrated initiative toward the decarbonization of road transportation[7,8].

Truck platooning emerges as a promising strategy to mitigate GHG emissions in road transportation[9,10]. This technique involves the cooperative and automated driving of multiple trucks. Real-time communication and collaboration enable a reduced following distance between trucks in a platoon. The following distance in a platoon could be reduced to at least 4 m[11], compared to the 30 m when driving individually[12]. This close-following driving mode effectively lowers air resistance[13], thereby reducing fuel consumption[14] and, consequently, leading to a nearly 10% reduction in GHG emissions[15]. Furthermore, through collaborative and close-proximity driving, truck platooning can mitigate congestion and traffic jams[16], thereby enhancing road capacity[17].

Despite its potential, the widespread commercial adoption of truck platooning still lacks motivation from logistics enterprises. Firstly, truck platooning's benefits from fuel consumption reduction are generally offset by additional costs, such as vehicle upgrade costs and labor costs. Truck platooning requires trucks to be upgraded by equipping advanced perception, communication, and computing devices[18]. Moreover, drivers are still required on each platooning truck, since there are no proven autonomous driving technologies capable of safely leading a platoon on open roads[19,20]. Secondly, the commercialization of truck platooning is concerned with safety risks, which may lead to serious public relations challenges[21,22]. Safety risks mainly derive from the ongoing debates surrounding self-driving

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technologies. The autonomous driving platoon may not handle all long-tail scenarios in reality.

To address these critical issues, recent research has explored human-lead truck platooning (HLTP), where a human-driven lead truck is followed by automated trucks[23,24]. These methods offer a dual advantage: they reduce the required number of human drivers to a single driver for the entire platoon, and critically, they leverage human perception and decision-making to handle long-tail scenarios. However, existing HLTP methods focus on longitudinal control but lack the lane-changing capability. This fundamental limitation poses a serious challenge: the inability to overtake slow-moving vehicles decreases logistical efficiency and thereby undermines the commercial operation of truck platooning.

To promote the commercial operation of truck platooning, we advocate for an approach to truck platooning, which has been implemented in the commercial operation on Shanghai Yangshan Port. Our proposed method incorporates the following key features aimed at addressing the current limitations, as shown in Fig. 1:

(a) With platoon lane-changing capability: in the actual implementation, the truck platoon may be impeded by slow-moving vehicles from time to time. To overcome this limitation, we propose a Human-Lead Truck Platooning with Lane-Changing Capability (HLTP-LC). The novel snake-like platoon lane-change strategy allows a truck platoon to efficiently change lane through a small gap.

(b) More cost-saving and efficient: besides enhancing ecology, the proposed HLTP-LC method also reduces vehicle upgrade costs and labor costs. In our design, the platoon leader is a Connected Human-driven Truck driven by a skilled human driver. Perception and computing devices do not need to be equipped on the platoon leader, since a skilled human driver is more reliable. In this HLTP-LC method, trucks behind the platoon leader do not need drivers anymore, thereby reducing labor costs. Furthermore, the platoon's lane-change ability allows for overtaking slow-moving vehicles, which increases logistical efficiency.

(c) Implemented in commercial operation: we have validated the proposed HLTP-LC method in a large-scale commercial operation at the Shanghai Yangshan Port, the world's largest port. Since the commercial operation, truck platooning has transported over 200,000 Twenty-foot Equivalent Units (TEU) annually.

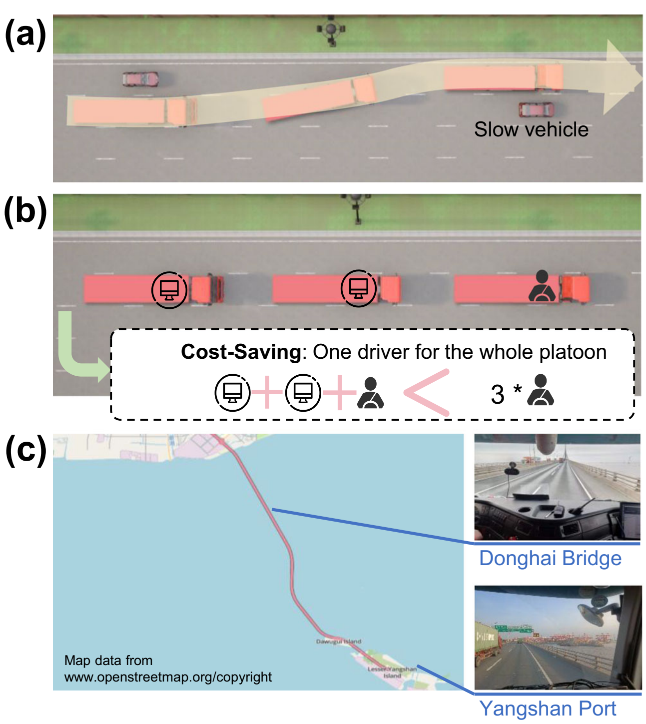


Fig. 1 | Key feature of the proposed HLTP-LC method. **a** with platoon lane-changing capability. **b** more cost-saving and efficient. **c** implemented in commercial operation.

Results

To validate the superiority of the proposed HLTP-LC method, we design four experiments, i) logistics efficiency evaluation; ii) sustainability performance in traffic iii) safety validation in long-tail scenarios; iv) field tests to validate sustainability and cost-saving. The results demonstrate that the proposed HLTP-LC method enhances logistics efficiency by increasing travel speed by 6.55%, when compared to the baseline platooning method. Moreover, compared with human-driven trucks, the proposed HLTP-LC method improves sustainability by reducing fuel consumption by 3.8% and GHG emissions by 5.1%. In terms of cost savings, it achieves a reduction in operating costs of 25.1% in the USA and 23.2% in China.

Logistics efficiency evaluation

Simulation platform. To evaluate the platooning logistics efficiency of the proposed HLTP-LC method in traffic, simulation tests are conducted using a platoon simulation platform. Built upon the Vissim traffic simulation software, the platform was custom-developed by our research team[25]. It can generate a sufficiently realistic road environment and traffic flow.

Test scenarios. The logistics route connecting the city of Shanghai to the Yangshan Deep-Water Port is remapped in the platoon simulation platform. In this scenario, the task of the truck platoon is to travel from Shanghai City to Yangshan Deep-Water Port. The simulated route is a six-lane, bidirectional highway with a speed limit of 80 km/h and features a predominantly straight geometry. Trucks account for 80% of the total traffic volume. Eight cases are designed with considering different traffic congestion levels, as shown in Supplementary TABLE I.

Three types of controllers are evaluated:

(1) Human-driven truck: a manually operated truck, serving as the real-world baseline without any automated functions.

(2) Baseline platooning method: it is a fully autonomous platooning method[26]. Without a human driver, it can not handle long-tail risks. Therefore, its operational design domain is intentionally restricted to longitudinal control to ensure safety.

(3) The proposed HLTP-LC method: by integrating longitudinal and lateral control, it can enhance truck platooning logistics efficiency by overtaking slow-moving vehicles. The parameter settings of the proposed HLTP-LC method are shown in Supplementary TABLE II.

Measures of effectiveness. Average travel speed is used as the measures of effectiveness of logistics efficiency. It is computed by dividing route distance by travel time.

Logistics efficiency results. Figure 2 illustrates the average travel speed of three controllers under varying traffic congestion levels. Compared to human-driven trucks, the proposed HLTP-LC method demonstrates an average increase in travel speed of 1.77%. However, when the volume/capacity ratio exceeds 0.75, the proposed HLTP-LC method performs slightly worse. This is primarily attributed to the insufficient space in adjacent lanes for truck platoons to overtake under high traffic congestion. Relative to the baseline platooning method, the proposed HLTP-LC method can achieve an average travel speed improvement of 6.55%, reaching a maximum value of 13.9% when the volume/capacity ratio is 0.125. However, the enhancement decreases as the volume/capacity ratio increases. This makes sense as the lane-change maneuver requires enough free space on the target lane. When the volume/capacity ratio increases, a suitable lane change space is difficult to find. Therefore, the proposed HLTP-LC can not easily execute lane-change maneuvers to overtake slow-moving vehicles. The performance of the HLTP-LC method thus closely approaches the baseline platooning method.

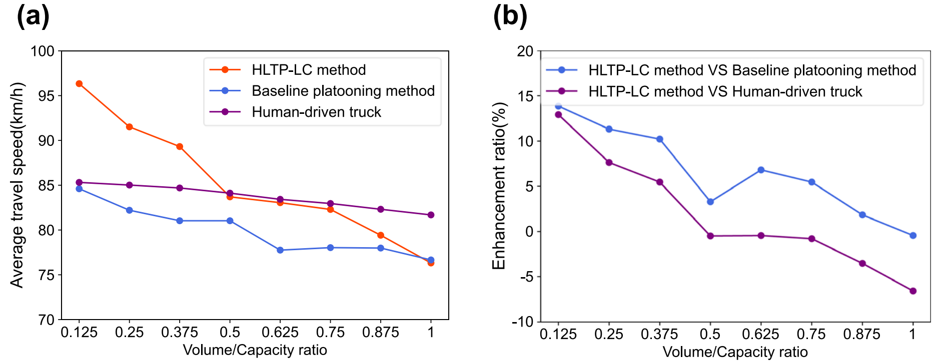


Fig. 2 | Platoon logistics efficiency results. **a** Average travel speed. **b** Enhancement ratio.

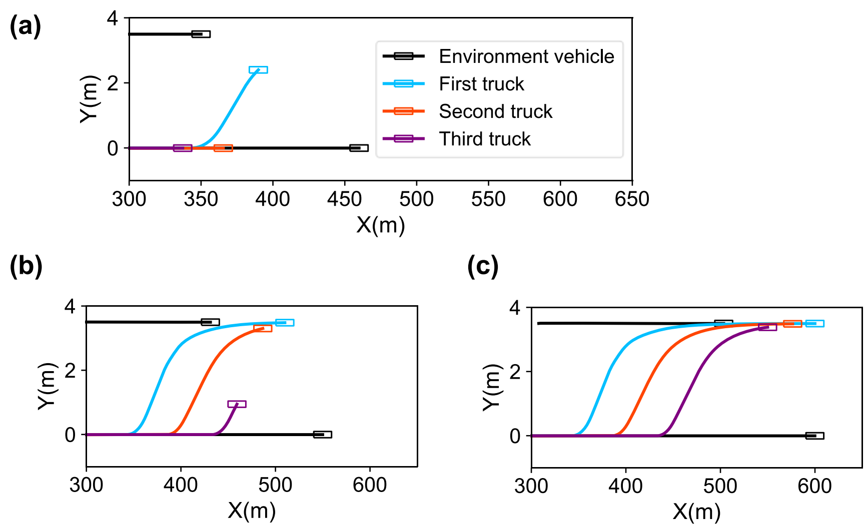


Fig. 3 | Platoon lane-change trajectory in simulation test. **a** Starting. **b** During. **c** Finishing.

The enhancement of logistics efficiency is attributed to the capability of overtaking slow-moving vehicles. We also conducted simulation and field tests to validate the overtaking capability, as illustrated by the snake-like lane-change trajectories shown in Fig. 3 and Fig. 4 (see Supplementary Movies 1 and 2). Figure 3 demonstrates a simulated three-trucks platoon to execute the lane-change maneuver. The successive lane-change maneuver of trucks can be observed. It confirms the capability of overtaking slow-moving vehicles. To validate this capability under real-world conditions, we conducted field tests, as shown in Supplementary Fig. 1. As presented in Fig. 4, a two-truck platooning successfully performed the maneuver, demonstrating the system's robustness in the presence of vehicle localization uncertainties.

Furthermore, when the platoon conducts an overtaking maneuver, it can maintain string stability[27], thus reducing its impact on traffic flow. As shown in Supplementary Fig. 2, the time gap between consecutive trucks is stable at different speeds. This is because the proposed HLTP-LC method can mitigate the impact of lateral lane changes on longitudinal control, thereby enhancing string stability. Moreover, the time gap error of second-third-truck is smaller than the time gap error of first-second-truck. It demonstrates that the string stability of the platoon increases with the increase in platoon scale. It also calls for the emergence of larger-scale platoon for logistics enterprises.

The proposed HLTP-LC method not only enhances the logistical efficiency of truck platoons but also improves road capacity by enabling trucks to drive collaboratively and in close proximity, thereby avoiding congestion and traffic jams[16]. This boosts overall road logistical efficiency. To validate this viewpoint, we conducted simulation experiments to evaluate road capacity under varying truck platoon penetration rates. Supplementary Fig. 3 illustrates an average increase of 18.95% in road capacity, with a maximum improvement of 38%. These findings are highly consistent with the related study[17], which reports a maximum increase of 42% for this scenario.

Sustainability performance in traffic

To validate the sustainability performance of the proposed HLTP-LC method in traffic flow environment, simulation tests are conducted across various traffic congestion levels, represented by the volume/capacity ratio. In these simulations, the proposed HLTP-LC method is benchmarked against human-driven trucks and the baseline platooning method[26]. Sustainability performance in traffic is quantified by estimating reductions in fuel consumption and GHG emissions across different cases. These two metrics are calculated based on simulation test data (see Supplementary Note 1 for the specific methodology).

As depicted in Fig. 5, the results indicate that the sustainability performance of the proposed HLTP-LC method is largely consistent with that of the baseline platooning method, with the HLTP-LC method's advantage being less than 1%. This suggests that existing truck platooning methods inherently offer sustainability benefits. However, when compared to human-driven trucks, the proposed HLTP-LC method achieves average reductions of 5.9% in fuel consumption and 7.6% in GHG emissions. Furthermore, these reductions are observed across nearly all levels of traffic congestion, demonstrating the superior sustainability performance of the proposed HLTP-LC method over human-driven trucks.

Safety validation in long-tail scenarios

The proposed HLTP-LC method is designed to be human-lead. It is capable of addressing long-tail scenarios where autonomous driving systems may

Fig. 4 | Platoon lane-change trajectory in field test. **a** Starting. **b** During. **c** Finishing.

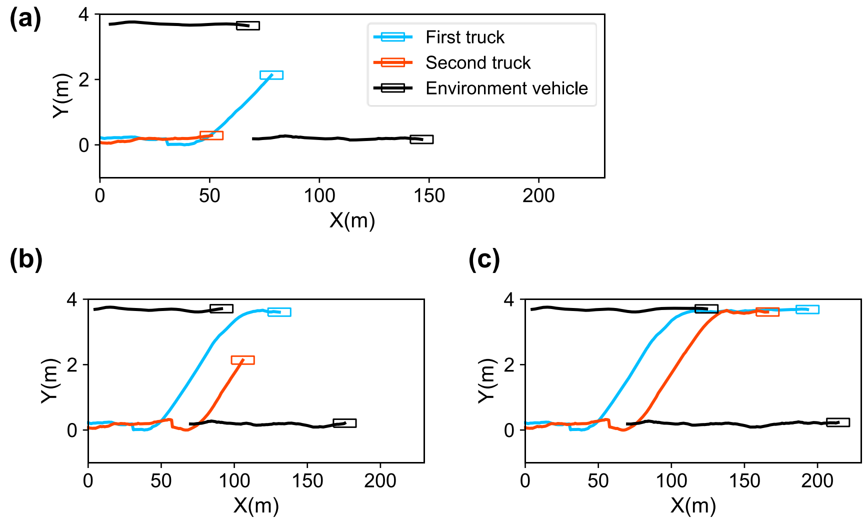
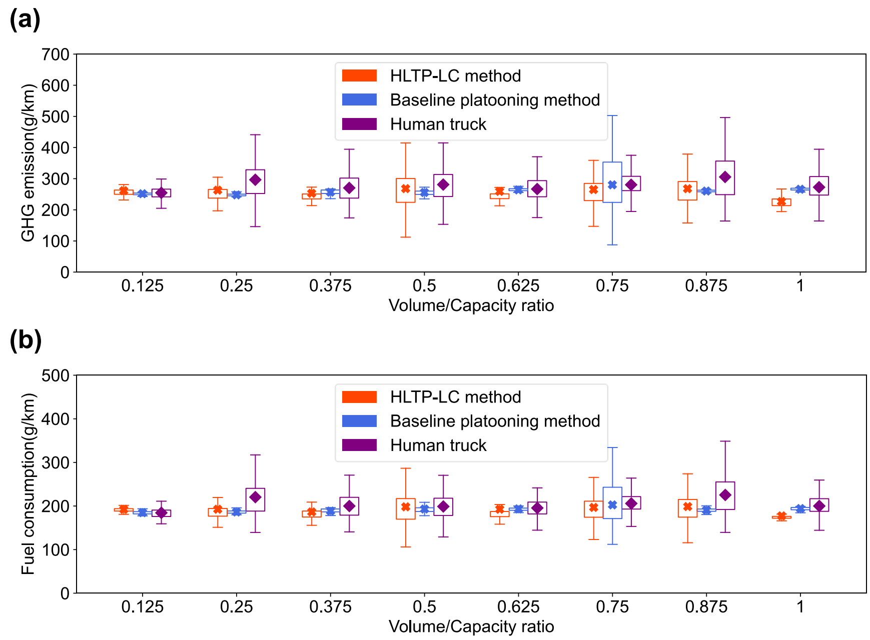


Fig. 5 | Performance of truck platooning under traffic flow. a GHG emissions. b fuel consumption.



falter[28]. Figure 6a illustrates a long-tail scenario: an obstacle falls from the front vehicle, which is similar to the driving environment. The truck platooning system fails to detect it. In this scenario, the first truck with a human driver can actively control the platoon to avoid the obstacle, while the following trucks can automatically follow the first truck's trajectory, as shown in Fig. 6. This design enhances the overall robustness and reliability of the truck platooning system when only one driver is used in the whole platoon.

Field tests to validate sustainability and cost-saving

Field Test Scenarios. To validate the sustainability and cost-saving of the proposed HLTP-LC method in the real world, we have conducted field tests on Yangshan Deep-Water Port, as illustrated in Fig. 1c. The port is recognized as the largest port globally. Using the proposed HLTP-LC method, a notable annual transport volume of 200,000 TEUs has been achieved.

Field Test Trucks. Designed by SAIC Motor, Hongyan Connected and Automated Truck is used for the truck platooning, as illustrated in Supplementary Fig. 4. The V2V communication of platooning relies on 5th Generation Mobile Communication Technology (5 G).

The Hongyan Connected and Automated Truck is a 6 axles tractor-semitrailer truck powered by a liquefied natural gas engine. It is equipped with a Nvidia Orin Sensor System to enable the real-time running of perception, decision-making and control algorithms. It is also equipped with a complicated perception system, as shown in Supplementary Fig. 5 and detailed in Supplementary TABLE III. The high-accuracy perception system lays the foundation for truck platooning.

Measures of Effectiveness. Measures of effectiveness of field test focus on sustainability and cost-saving. Sustainability performance is measured through estimated reductions in fuel consumption and GHG emissions, when compared to human-driven trucks. They are calculated from field test data (see Supplementary Note 1 for the specific methodology).

Concurrently, the cost-saving is determined by estimating the decrease in operational costs. This involves a comprehensive comparison of fuel, management, and labor costs against the costs associated with both human-driven trucks and a fully autonomous truck platooning. Moreover, to

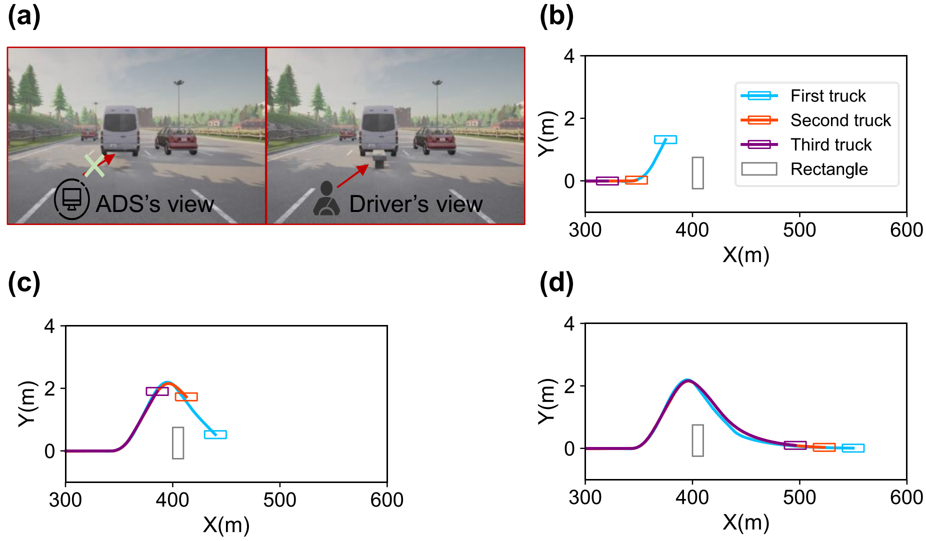


Fig. 6 | Trajectory in long-tail scenarios. **a** The autonomous driving system (ADS) fails to detect a fallen obstacle, whereas a human driver detects it. **b** Starting. **c** During. **d** Finishing.

evaluate the cost-saving in different regions, the operating costs for China and the United States are presented in the following Supplementary TABLE IV. The operating costs for the United States is based on publicly available data[29,30], while the operating costs for China is derived from the commercialization operation conducted by the research team's affiliated company.

Sustainability performance. For fuel consumption, results indicate reductions per truck can reach 2.5% and 3.8%, respectively, for the 2-trucks and 3-trucks platooning system. There are two reasons why the proposed HLTP-LC method can reduce fuel consumption. The first reason is that the speed fluctuations of the following trucks can be smoothed, through cooperative control of trucks in platoon, as shown in Fig. 7a. Therefore, unnecessary acceleration and deceleration behaviors of second and third trucks are reduced in Fig. 7b. It leads to reductions in instantaneous and cumulative fuel consumption in Fig. 7c and d. The second reason is that the smaller gap between trucks in the platoon reduces air drag for the following trucks (excluding the platoon leader). Therefore, fuel consumption per truck would decrease. As shown in Fig. 7e, an interesting phenomenon is that the fuel consumption distribution of the further back truck is closer to zero. Therefore, it can be concluded that as the number of trucks in a platoon increases, the average fuel consumption per truck will decrease. This phenomenon calls for the emergence of larger-scale platoons for logistics enterprises.

For GHG emissions, results indicate reductions per truck can reach 3.6% and 5.1% respectively, for the 2-trucks and 3-trucks platooning system. It is apparent that the reduction in GHG emissions primarily comes from the decrease in total fuel consumption. However, the reduction in GHG emissions is greater than fuel consumption. This is because the proposed HLTP-LC method reduces the trucks' abrupt acceleration and deceleration behaviors, as shown in Fig. 7a. Consequently, fuel can be more efficiently burned and fully utilized, further reducing GHG emissions. Taking the example of a 3-trucks platooning system, Fig. 7f, g respectively illustrate the instantaneous and cumulative GHG emissions of the truck platoon. It can be observed that the trucks further back in the platoon experience lower GHG emissions, compared to the front trucks. Therefore, a larger-scale platoon can bring about lower average GHG emissions per truck.

Cost-Saving estimation. In the USA, the proposed HLTP-LC method can save logistics enterprises 25.1% in operating costs compared to human-driven trucks. The cost distribution for human-driven trucks, as shown in Fig. 8a, includes management, labor, and fuel costs. By utilizing the proposed HLTP-LC method, fuel costs can be reduced by approximately 3%. Moreover, only the platoon leader requires a driver. Taking a 3-truck platoon as an example, this results in a two-thirds reduction in labor costs, as depicted in Fig. 8b. Furthermore, Fig. 8c illustrates the advantages of fully autonomous truck platoon compared to human-driven trucks. To ensure safe platoon operation, fully autonomous platoons require a safety driver in each truck, thus negating the labor cost advantage and offering only a marginal fuel cost advantage, accounting for just 0.4% of the total cost.

To evaluate cost-saving in different regions, we analyze truck platooning costs in China. Unlike in the USA, labor and management costs constitute a lower proportion of total costs in China, while fuel costs are higher. Overall, compared to human-driven trucks, as shown in Fig. 8d, the proposed HLTP-LC method can save logistics enterprises 23.2% in operating costs, as shown in Fig. 8e. Fully autonomous platoons can achieve 1.8% in operating cost savings for logistics enterprises, as shown in Fig. 8f.

However, the truck's upgrade for truck platooning systems requires additional upgrade costs, which logistics enterprises must bear. Even with this cost, the proposed HLTP-LC method demonstrates a short payback period, as illustrated in Supplementary Fig. 6. Based on the assumption that all trucks operate under the HLTP-LC method and that the total annual operating mileage remains the same, the payback period in the United States is approximately 0.35 years, as the savings in labor and fuel costs far outweigh the upgrade cost. In China, the payback period extends to 1.12 years. This longer period is primarily due to the lower overall truck operating costs in China, resulting in comparatively smaller cost savings and consequently a longer payback period. Overall, the proposed HLTP-LC method remains highly attractive to logistics enterprises and exhibits strong commercial prospects.

Discussion

The proposed HLTP-LC method addresses the commercialization challenge of current truck platooning technology via the following contributions. Firstly, the proposed method enhances ecological benefits, reducing fuel consumption by 3.8% and GHG emissions by 5.1%. As the number of trucks in a platoon increases, the ecological benefits scale accordingly. Secondly, the proposed method ensures implementation safety and reduces logistics enterprise's costs by 25.1% in the USA and 23.3% in China, respectively. Long-tail risks are greatly avoided by utilizing a human-driven truck to guide the whole platoon. Additionally, vehicle upgrade costs are reduced since the platoon leader no longer requires advanced perception devices. Labor costs are reduced by using only one driver in a platoon. Truck

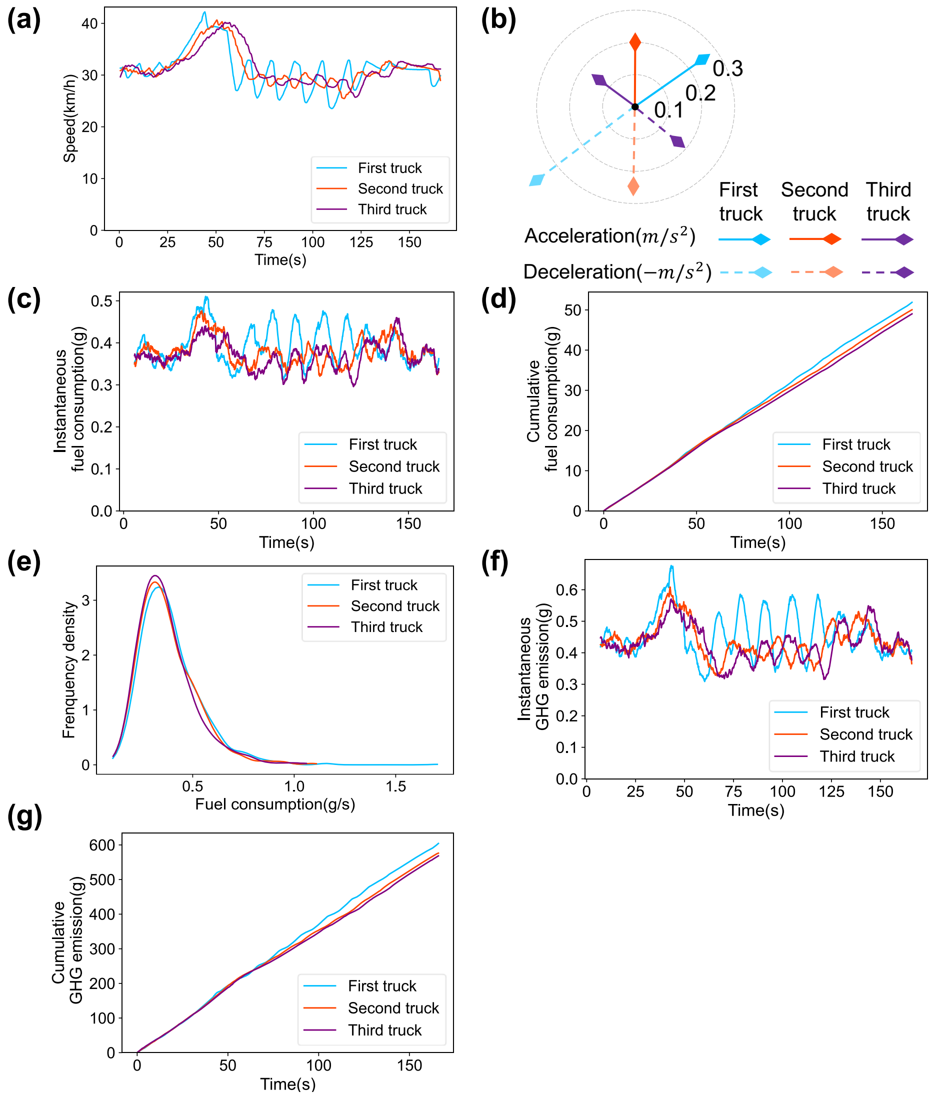


Fig. 7 | Fuel consumption and GHG emissions of trucks in platoon. **a** Speed profile of trucks. **b** Acceleration range of trucks. **c** Instantaneous fuel consumption. **d** Cumulative fuel consumption. **e** Distribution of fuel consumption. The y-axis denotes the frequency density, defined as frequency divided by the width of each fuel-consumption interval. **f** Instantaneous GHG emissions. **g** CUmulative GHG emissions.

platooning's payback period is reduced to only 0.35 years in the USA and 1.12 years in China. Thirdly, the lane-changing capability improves logistics efficiency by 6.55% through overtaking slow-moving vehicles, when compared to the baseline platooning method.

Although the proposed HLTP-LC method has addressed most of the challenges in the commercialization of truck platooning, several critical aspects require further consideration before its widespread application. Regarding policy and regulatory implications, the current commercial operation of our truck platooning system is legally supported within government-approved autonomous driving test roads. However, the broader adoption and scaling of truck platooning, particularly in other regions, will necessitate a more mature and comprehensive regulatory landscape. To address this, we can work closely with policymakers to help shape supportive laws, and advocate for consistent rules across different regions.

Furthermore, operational challenges under adverse weather conditions present a serious limitation. While the human-lead truck platooning mitigates some risks, dense fog, heavy rain, or icy conditions can severely impair perception and communication, posing threats to closely spaced trucks. Future work should explore dynamically increasing inter-vehicle spacing during bad weather to enhance safety.

Finally, cybersecurity risk is a paramount concern. The communication system vital for truck platooning is vulnerable to cyber-attacks like jamming or data manipulation, potentially causing unsafe vehicle behavior or system takeovers. Future work must prioritize developing robust, multi-layered cybersecurity protocols, including secure channels and fail-safe mechanisms. Addressing this is crucial for safe commercial deployment.

Methods

In the proposed HLTP-LC method, the platoon leader is operated by a human driver, while the following trucks are controlled by the platooning system. Typically, the platoon travels in a single lane. When the system determines that a lane change is necessary, it prompts the human driver to maneuver the platoon leader change lane. The following trucks then autonomously execute the lane change maneuver under the control of the platooning system.

The control flowchart of the proposed HLTP-LC method, as illustrated in Fig. 9, is comprised of three distinct modules:

Module 1: longitudinal gap regulation controller. This module is responsible for the longitudinal control of the following trucks. As shown in Fig. 9, it continuously monitors the platoon's state and computes the state error relative to other trucks. Based on the state error, it computes the optimal acceleration to maintain a stable time gap.

Module 2: lane-change decision-making algorithm. This module serves as the high-level decision-maker for lane-change maneuvers. It activates when a lane change might be beneficial, either for rerouting

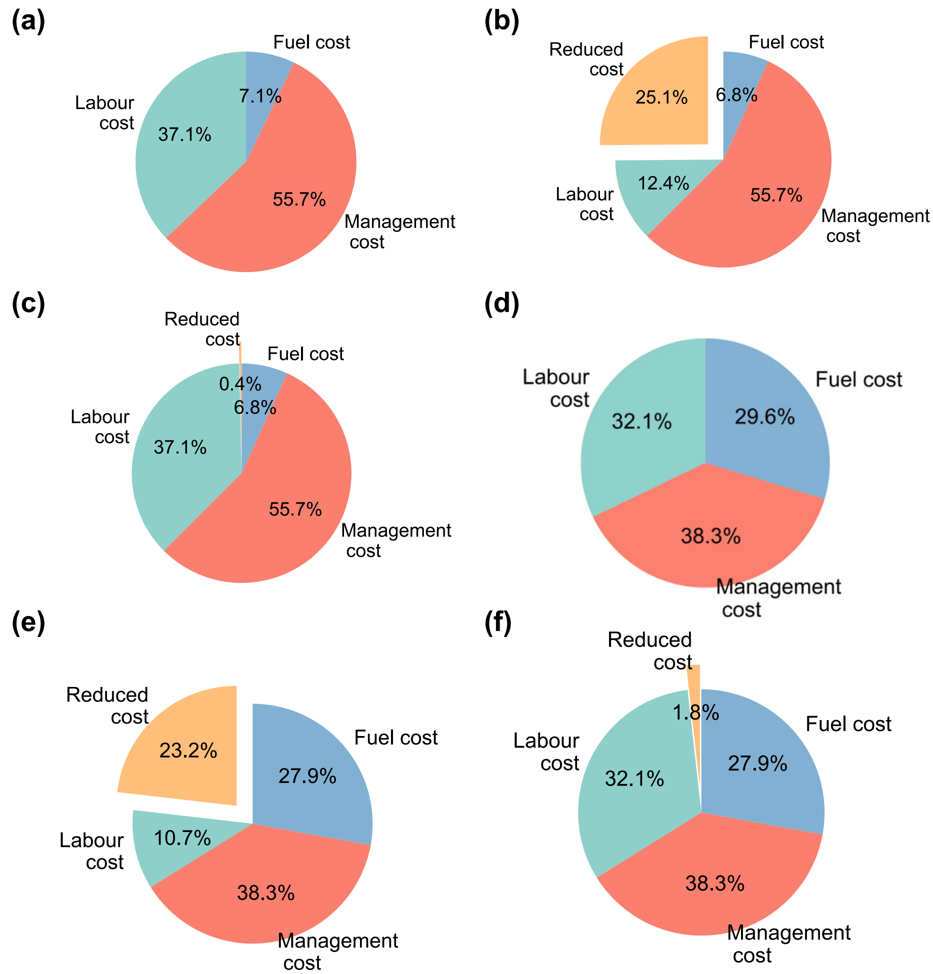


Fig. 8 | Cost comparison. a human-driven truck in USA. **b** human-lead truck platoon in USA.**c** fully autonomous platoon in USA. **d** human-driven truck in China. **e** human-lead truck platoon in China. **f** fully autonomous platoon in China.

purposes or to overtake slower vehicles impeding the platoon's logistics efficiency. When the lane change maneuver is safe, the algorithm outputs a lane-change request. This request serves two functions simultaneously: it prompts the human driver to maneuver the platoon leader change lane. and it signals Module 3 for lateral control of the following trucks.

Module 3: lateral controller. This module executes the lateral lane-change control for the following trucks. Based on the lane change trajectory of the leading truck, it computes the optimal front wheel angle for its own truck to follow this trajectory. This ensures that each following truck replicates the leader's trajectory, enabling the truck platoon change lanes by turning like a snake.

Longitudinal gap regulation controller

To control the following truck's longitudinal movement, a gap regulation controller[31] is proposed. Based on the model predictive control method, the controller aims to maintain a desired time gap between the following truck and the leading truck.

System Dynamics. The conventional gap regulation controller is formulated within the time-domain, where its function is to preserve a desired distance from the leading truck. A challenge with this formulation is that a minor speed fluctuation from the leading truck can swiftly alter the actual distance, compelling the following truck to change speed immediately. It often results in a greater speed fluctuation for the following truck. To avoid the problem, the system dynamics of the proposed gap regulation controller are formulated with respect to distance instead of time. Consequently, the system dynamics are space-domain based. Variables used in space domain are shown in Supplementary TABLE V.

Definition 1: By taking the first-order derivative of time t in relation to position s, we define slowness w as follows:

$$w = \frac{dt}{ds} \tag{1}$$

Definition 2: By taking the second-order derivative of time t in relation to position s, we define moderation b as follows:

$$b = -\frac{a}{v^3} \tag{2}$$

where a is the acceleration.v is the speed.

Using the space-domain variables presented above, we can define truck state vector $\boldsymbol{\xi}$ and control vector $\mathbf{u}$ are defined as follows:

$$\boldsymbol{\xi} \overset{\text{def}}{=} \left[g_l^* - g_l, w_l - w, g_p^* - g_p, w_p - w, b\right]^T \tag{3}$$

$$\mathbf{u} \overset{\text{def}}{=} \Delta b \tag{4}$$

where g_l^* and g_l represent the desired and actual time gap between ego truck and leading truck respectively, considering their interaction. Similarly, g_p^* and g_p are the desired and actual time gap between ego truck and preceding truck. w_l denotes ego truck's slowness. w_p is preceding truck's slowness. b is the ego truck's moderation. Δb is the increment of b.

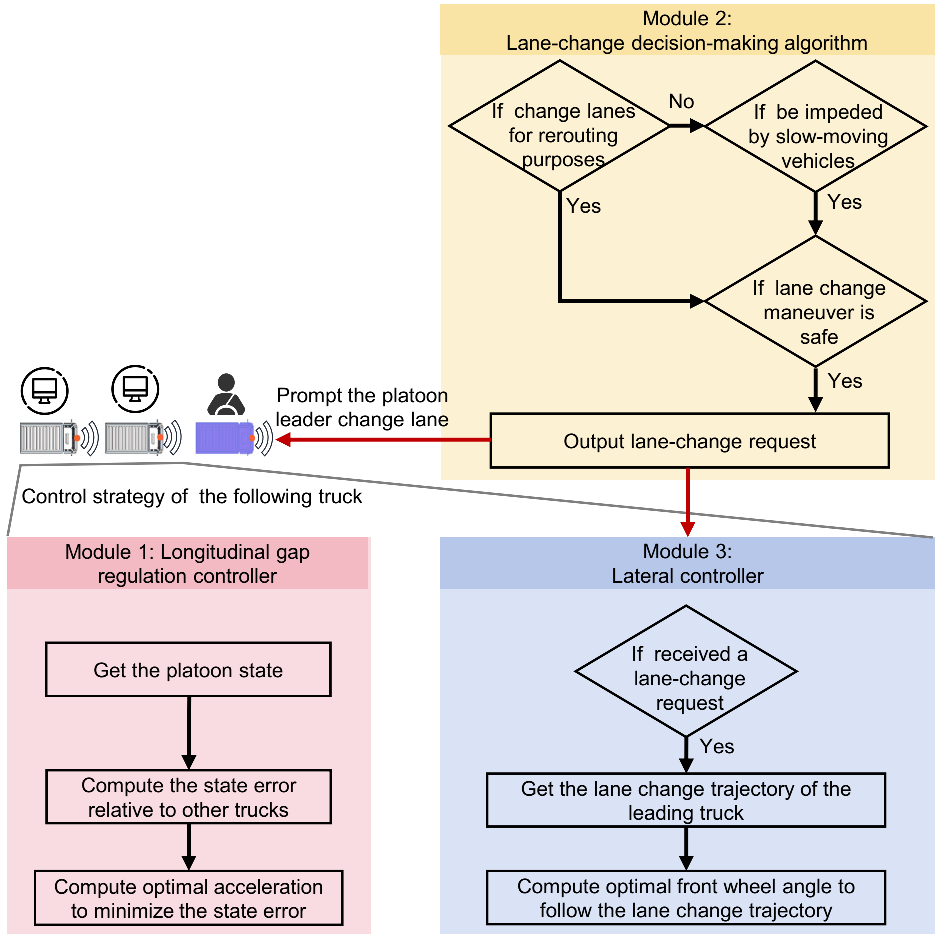


Fig. 9 | Control flowchart of the proposed HLTP-LC method. The flowchart consists of three modules. Module 1 regulates the longitudinal gap of each following truck. Module 2 determines whether the platoon should output a lane-change request. Module 3 executes the lateral lane-change control for the following trucks.

The longitudinal vehicle dynamics can be formulated by adopting a particle motion model as follows:

$$\mathrm{\xi_{i+1} = A_i\xi_i + B_i u_i + C_i\omega_i} \tag{5}$$

$$\mathbf{A_i} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \times \Delta t + \mathbf{I}_{5\times 5} \tag{6}$$

$$\mathbf{B_i} = \begin{bmatrix} 0 \\ -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} \times \Delta t \tag{7}$$

$$\mathbf{C_i} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \times \Delta t \tag{8}$$

$$\boldsymbol{\omega}_\mathbf{i} = \begin{bmatrix} b_l & b_p \end{bmatrix} \tag{9}$$

where b_l is the moderation of leading truck; b_p is the moderation of preceding truck.

Two implementation issues arise from the cubic relation between moderation and speed in Eq. (2): the difficulty of finding a suitable weight factor in the cost function for all speed conditions, and the unavoidable singularities that occur as very low speeds cause b to become infinite.

The proposed solution, shown in Eq. (10), is to transform the control vector from the increment of moderation to acceleration.

$$\boldsymbol{u}' \overset{\text{def}}{=} a \tag{10}$$

By combining Eqs. (2)-(9), the final vehicle dynamics is presented as follows.

$$\mathrm{\xi_{i+1} = A'_i\xi_i + B'_i u'_i + C'_i\omega'_i} \tag{11}$$

$$\mathrm{A'_i = A_i} \tag{12}$$

$$\mathbf{B'_i} = \begin{bmatrix} 0 \\ w^3 \\ 0 \\ w^3 \\ 1 \end{bmatrix} \times \Delta t \tag{13}$$

$$\mathbf{C'_i} = \begin{bmatrix} 0 & 0 \\ -w_l{}^3 & 0 \\ 0 & 0 \\ 0 & -w_p{}^3 \\ 0 & 0 \end{bmatrix} \times \Delta t \tag{14}$$

$$\boldsymbol{\omega'}_\mathbf{i} = \begin{bmatrix} a_l & a_p \end{bmatrix} \tag{15}$$

where a_l is the acceleration of leading truck; a_p is the acceleration of preceding truck.

Cost function. To accelerate computation, cost functions is formulated into a quadratic form. The cost function is formulated as follows:

$$\min_{u'} \left(\sum_{i=1}^{N} \left(\underbrace{Q||\xi_i||_2}_{\text{State error cost}} + \underbrace{R||u'_i||_2}_{\text{Control cost}} \right) + \underbrace{Q||\xi_{N+1}||_2}_{\text{Terminal state error cost}} \right) \tag{16}$$

$$\mathbf{Q} = \begin{bmatrix} \beta_1 & 0 & 0 & 0 & 0 \\ 0 & \beta_2 & 0 & 0 & 0 \\ 0 & 0 & \beta_3 & 0 & 0 \\ 0 & 0 & 0 & \beta_4 & 0 \\ 0 & 0 & 0 & 0 & \beta_5 \end{bmatrix} \tag{17}$$

where state error cost **Q** is a positive-definite matrix; control effort cost **R** is a non-negative constant.

Constraints. Ego truck must satisfy the following constraints.

$$w_{\min} \le w \le w_{\max} \tag{18}$$

$$a_{\min} \le a \le a_{\max} \tag{19}$$

where $w_{\min}$ and $w_{\max}$ are the minimum and maximum slowness respectively; $a_{\min}$ and $a_{\max}$ are the minimum and maximum acceleration respectively.

Solution method. At each time step, the longitudinal controller aims to find an optimal sequence of accelerations over the control horizons, while satisfying the truck's system dynamics and constraints. The longitudinal controller is solved by a Dynamic-Programming based model predictive control algorithm, which leverages the structure of dynamic programming. Instead of tackling the entire sequence of future accelerations as a complex optimization variable, the algorithm transforms the multi-stage optimization into a series of simpler, sequential sub-problems. By working backward from the end of the control horizon, it determines the optimal action for each discrete time step, building the complete optimal acceleration sequence one step at a time. This stage-wise decomposition reduces the computational complexity compared to methods that optimize all variables simultaneously. The high efficiency of the algorithm ensures the truck can react swiftly to changing traffic conditions. The detailed solving algorithm is presented in our previous study[32].

Lane-Change decision-making algorithm

This section describes our lane change decision-making algorithm, which serves as a high-level module that determines whether a lane change is safe. As shown in Fig. 9, the algorithm is activated when the platoon needs to overtake slower vehicles or change lanes for rerouting. It assesses the safety of lane-change maneuver by comparing real-time distances between the platoon and surrounding vehicles against pre-computed minimum safe distances. If the maneuver is safe, it outputs a lane-change request. Details of the maneuver safety assessment algorithm are presented in Table 1.

In Table 1, $d_{RVT,PVT}$ is the distance of RVT and PVT; $d_{RVT,LT}$ is the distance of RVT and LT; $d_{LT,PVT}$ is the distance of LT and PVT; v_{RVT}, v_{PVT}, v_{LT}, v_{PV} is the speed of RVT, PVT, LT and PV respectively; $v_{P,desire}$ is the desired speed of the truck platooning; ω is a parameter defined by the user, such as logistics enterprise.

$d^{\min}_{RVT,PVT}$, $d^{\min}_{RVT,LT}$, $d^{\min}_{LT,PVT}$ are the minimum safe distance. They can be obtained by considering reaction behavior of human divers and the maximum braking capacity of vehicles.

$d^{\min}_{RVT,LT}$ can be obtained as Eq. (20).

$$d^{\min}_{RVT,LT} = \left(t^1_{RVT} + t^2_{RVT}\right)v_{RVT} + \frac{{v_{RVT}}^2}{2a_{RVT\min}} - \frac{{v_{LT}}^2}{2a_{LT\min}} + l \tag{20}$$

where t^1_{RVT} is the driver's reaction time of RVT; t^2_{RVT} is brake's response time of RVT; $a_{RVT\min}$ and $a_{LT\min}$ are minimum acceleration of RVT and LT respectively; l is the additional safe distance.

$d^{\min}_{LT,PVT}$ can be obtained as Eq. (21),

$$d^{\min}_{LT,PVT} = \left(t^1_{LT} + t^2_{LT}\right)v_{LT} + \frac{{v_{LT}}^2}{2a_{LT\min}} - \frac{{v_{PVT}}^2}{2a_{PVT\min}} + l \tag{21}$$

where t^1_{LT} and t^2_{LT} is the driver's reaction time and brake's response time of LT; $a_{PVT\min}$ is minimum acceleration of PVT.

$d^{\min}_{RVT,PVT}$ can be obtained as Eq. (22).

$$d^{\min}_{RVT,PVT} = d^{\min}_{RVT,LT} + L_{platoon} + d^{\min}_{LT,PVT} \tag{22}$$

where $L_{platoon}$ is platoon's length.

Lateral controller

In this section, a lateral controller is proposed that activates in response to the lane change request. This controller enables the following trucks to execute a lane change maneuver at fixed or moving position by turns like a snake. Supplementary Fig. 7 illustrates the scenario of lane change at a moving position. The controller is formulated as an optimal control problem, allowing for efficient solving computation.

Control Objective. To allow the following truck to follow the lane change trajectory of the leading truck, the objective is defined in the following equation.

$$y(x^r) - y^0(x^r) = 0 \tag{23}$$

where x^r is the relative longitudinal position with respect to the reference object in the frenet coordinates; $y^0(x^r)$ is the lateral position of leading truck at the relative longitudinal position x^r; $y(x^r)$ is the lateral position of following truck at relative longitudinal position x^r.

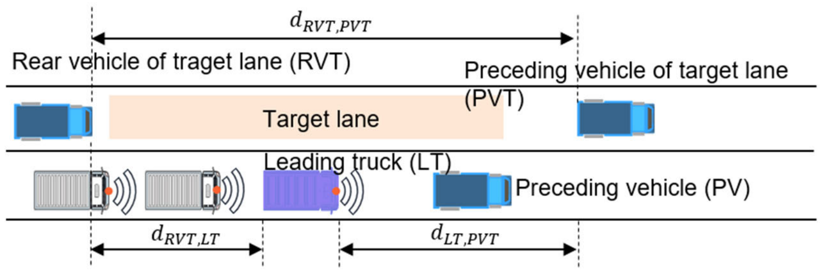
States and control variables. Different from non-articulated vehicles, a tractor-semitrailer truck needs more variables to depict. State variable $\boldsymbol{\xi}_{\mathbf{lat}}$ and control variable $\mathbf{u}_{\mathbf{lat}}$ are defined as follows.

$$\boldsymbol{\xi}_{\mathbf{lat}} \overset{\text{def}}{=} \left[y, e_\varphi, \epsilon_f, \dot{y}, \dot{e_\varphi}, \dot{\epsilon_f}\right]^T \tag{24}$$

$$\mathbf{u}_{\mathbf{lat}} \overset{\text{def}}{=} \delta_f \tag{25}$$

as shown in Supplementary Fig. 8, y is the lateral position of center of gravity of tractor; e_φ is the heading angle of tractor; ϵ_f is the angle of tractor respect to semitrailer; δ_f is the front wheel angle of tractor.

Table 1 | Platoon lane change maneuver safety assessment algorithm



Input: $d_{RVT,PVT}$; $d_{RVT,LT}$; $d_{LT,PVT}$; v_{RVT}; v_{PVT}; v_{LT}; v_{PV};$v_{P,desire}$

Output: Lane change request

Compute minimum safe distance $d^{min}_{RVT,PVT}$, $d^{min}_{RVT,LT}$, $d^{min}_{LT,PVT}$ as Eqs. (20)-(22)

If $d_{RVT,PVT} > d^{min}_{RVT,PVT}$ **then**

  If $d_{RVT,LT} > d^{min}_{RVT,LT}$ **then**

    If $d_{LT,PVT} > d^{min}_{LT,PVT}$ **then**

      Output Lane change request

    End if

  End if

End if

System Dynamics. The lateral vehicle dynamics can be formulated by adopting a Lagrange Mechanics-based tractor-semitrailer model[25,33] as follows:

$$\xi_{lat i+1} = A_i \xi_{lat i} + B_i u_{lat i} + C_i \tag{26}$$

$$A_{lat,i} = \begin{bmatrix} -ND & -NK \\ I_{3\times3} & 0_{3\times3} \end{bmatrix} \times \Delta t + I_{6\times6} \tag{27}$$

$$B_{lat,i} = \begin{bmatrix} NF \\ 0_{3\times1} \end{bmatrix} \times \Delta t \tag{28}$$

$$C_{lat,i} = \begin{bmatrix} NE_1 \dot{e}_d + NE_2 \ddot{e}_d \\ 0_{3\times1} \end{bmatrix} \times \Delta t \tag{29}$$

where $\dot{e}_d$ is the desired yaw rate set by the road, $\ddot{e}_d$ is the desired yaw acceleration set by the road, Δt is the control time step size, **N** is the inverse of the inertial matrix, **D** is the damping matrix, **K** is the potential matrix, **F** is the vector representing the tire cornering stiffness, $\mathbf{E_1}$ and $\mathbf{E_2}$ are matrices representing the disturbance effects of $\dot{e}_d$ and $\ddot{e}_d$, respectively, they are defined in the previous research.

Cost Function. The lateral controller plans $N + 1$ steps of motions with step interval Δt. In i_{th} step, the desired lateral state vector ξ_{lat,des_i} is defined as follows.

$$\xi_{lat,des_i} \overset{def}{=} [\Omega(x_i), 0, 0, 0, 0, 0]^T \tag{30}$$

where x_i is the planned longitudinal position of i_{th} step in the longitudinal controller. Ω records the historical trajectory of the leading truck. $\Omega(x_i)$ donates the lateral position of the leading truck when the relative longitudinal position is x_i.

Considering state cost and control const in all predictive horizons, the cost function is formulated as follows:

$$\min_u \left(\sum_{i=1}^{N} \left(\underbrace{Q_{lat} ||\xi_{lat i} - \xi_{lat,des_i}||_2}_{\text{State error cost}} + \underbrace{R_{lat} u_{lati_2}}_{\text{Control cost}} \right) + \underbrace{Q_{lat} \xi_{lat N+1} - \xi_{lat,des_{N+1}2}}_{\text{Terminal state error cost}} \right) \tag{31}$$

$$\mathbf{Q_{lat}} = \begin{bmatrix} \beta_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \beta_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \beta_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & \beta_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & \beta_5 & 0 \\ 0 & 0 & 0 & 0 & 0 & \beta_6 \end{bmatrix} \tag{32}$$

where state error cost $\mathbf{Q}_{\mathbf{lat}}$ is a positive-definite matrix; control effort cost $\mathbf{R}_{\mathbf{lat}}$ is non-negative.

Constraints. The following constraints must be satisfied.

$$\epsilon_{f\,\min} \le \epsilon_f \le \epsilon_{f\,\max} \tag{33}$$

$$\delta_{f\,\min} \le \delta_f \le \delta_{f\,\max} \tag{34}$$

where $\epsilon_{f\,\min}$ and $\epsilon_{f\,\max}$ are the minimum and maximum angle of tractor respect to semitrailer respectively; $\delta_{f\,\min}$ and $\delta_{f\,\max}$ are the minimum and maximum front wheel angle respectively.

Solution method. The lateral control problem of truck platooning is also solved by the Dynamic-Programming based model predictive control algorithm, following the same solution method as that adopted for the longitudinal gap regulation controller.

Safety and Fault Tolerance Mechanism

In addition to the three primary control modules, a safety and fault tolerance mechanism is integrated into the truck platooning system to ensure operational robustness and reliability. This mechanism is designed to detect, assess, and mitigate potential system failures, ensuring the platoon always operates in a safe state. The key failures consist of communication failure, onboard system failure, and cybersecurity risks.

(1) Communication Failure: the V2V communication is the backbone of the platoon. To handle potential communication failure, such as signal loss and high latency, a continuous message is exchanged between all trucks. If a following truck fails to receive this message from other trucks for a predefined time, it triggers the communication failure procedure. The truck will immediately disengage from the platoon and transition to a standard Adaptive Cruise Control mode.

(2) Onboard System Failure: the following trucks rely on critical onboard systems, such as radars and cameras. A built-in diagnostic system continuously monitors the health of these onboard systems. If a critical fault is detected, the affected truck broadcasts an emergency message to the platoon, initiates a controlled deceleration, and executes a safe-stop maneuver.

(3) Cybersecurity Risks: given the reliance on wireless communication, the truck platooning system is designed with cybersecurity in mind. All V2V messages are encrypted and authenticated to prevent unauthorized access or malicious attacks. A cybersecurity intrusion detection system monitors cybersecurity risks. If a credible risk is detected, the platoon is immediately dissolved. All following trucks will revert to independent operation in ACC mode.

Data availability

The datasets during the current study are available from the corresponding author on reasonable request.

Code availability

The code during the current study are available from the corresponding author on reasonable request.

Received: 17 March 2025; Accepted: 18 December 2025;
Published online: 07 January 2026

References

1. Meinshausen, M. et al. Greenhouse-gas emission targets for limiting global warming to 2 C. *Nature* **458**, 1158–1162 (2009).
2. Kweku, D. W. et al. Greenhouse effect: greenhouse gases and their impact on global warming. *J. Sci. Res. Rep.* **17**, 1–9 (2018).
3. Montzka, S. A., Dlugokencky, E. J. & Butler, J. H. Non-CO2 greenhouse gases and climate change. *Nature* **476**, 43–50 (2011).
4. *U.S. energy statistics*, <https://www.eia.gov/totalenergy/data/monthly> (UIA, 2025).
5. Aminzadegan, S., Shahriari, M., Mehranfar, F. & Abramović, B. Factors affecting the emission of pollutants in different types of transportation: a literature review. *Energy Rep.* **8**, 2508–2529 (2022).
6. Holmberg, K., Andersson, P. & Erdemir, A. Global energy consumption due to friction in passenger cars. *Tribology Int.* **47**, 221–234 (2012).
7. Zhang, R. & Hanaoka, T. Cross-cutting scenarios and strategies for designing decarbonization pathways in the transport sector toward carbon neutrality. *Nat. Commun.* **13**, 3629 (2022).
8. Zhang, S. & Chen, W. Assessing the energy transition in China towards carbon neutrality with a probabilistic framework. *Nat. Commun.* **13**, 1–15 (2022).
9. Paddeu, D. & Denby, J. Decarbonising road freight: is truck automation and platooning an opportunity? *Clean Technol. Environ. Policy* 1−15 (Springer, 2021).
10. Stasinopoulos, P., Shiwakoti, N. & Beining, M. Use-stage life cycle greenhouse gas emissions of the transition to an autonomous vehicle fleet: A System Dynamics approach. *J. Clean. Prod.* **278**, 123447 (2021).
11. Lu, X.-Y. & Shladover, S. E. Automated truck platoon control and field test. *Road Vehicle Automation*, 247-261 (Springer, 2014).
12. Brackstone, M., Sultan, B. & McDonald, M. Motorway driver behaviour: studies on car following. *Transp. Res. Part F Traffic Psychol. Behav.* **5**, 31–46 (2002).
13. Browand, F., Zabat, M. & Tokumaru, P. Aerodynamic benefits from close-following. *Automated Highway Systems*, 247-264 (Springer, 1997).
14. Michaelian, M. & Browand, F. Field experiments demonstrate fuel savings for close-following. (Springer, 2000).
15. Cheng, H. et al. Truck platooning reshapes greenhouse gas emissions of the integrated vehicle-road infrastructure system. *Nat. Commun.* **14**, 4495 (2023).
16. Guerrieri, M. & Pugno, N. ANTi-JAM solutions for smart roads: ant-inspired traffic flow rules under CAVs environment. *Transp. Res. Interdiscip. Perspect.* **29**, 101331 (2025).
17. Guerrieri, M. Smart roads geometric design criteria and capacity estimation based on AV and CAV emerging technologies. A case study in the trans-European transport network. *Int. J. Intell. Transp. Syst. Res.* **19**, 429–440 (2021).
18. Marzano, V. et al. Impacts of truck platooning on the multimodal freight transport market: an exploratory assessment on a case study in Italy. *Transp. Res. Part A Policy Pract.* **163**, 100–125 (2022).
19. Tsugawa, S., Jeschke, S. & Shladover, S. E. A review of truck platooning projects for energy savings. *IEEE Trans. Intell. Veh.* **1**, 68–77 (2016).
20. Janssen, R., Zwijnenberg, H., Blankers, I. & De Kruijff, J. Truck platooning. *Driving the* (IEEE, 2015).
21. Windover, P., Owens, R. & Roy, B. Truck Platooning Policy Barriers Study. (New York State Energy Research and Development Authority, 2018).
22. Engström, J. et al. Deployment of automated trucking: challenges and opportunities. Road Veh. Autom. **5**, 149–162 (2019).
23. Zhang, Y. et al. Human-lead-platooning cooperative adaptive cruise control. *IEEE Trans. Intell. Transp. Syst.* **23**, 18253–18272 (2022).
24. Hu, J. et al. Safety-aware human-lead vehicle platooning by proactively reacting to uncertain human behaving. *Transp. Res. Part C Emerg. Technol.* **170**, 104941 (2025).
25. Hu, J. et al. A Simulation platform for truck platooning evaluation in an interactive traffic environment. *IEEE Trans. Intell. Transp. Syst.* **25**, 13785–13804 (2024).
26. Li, Y. et al. A hierarchical controller for connected truck platoon: analysis and verification. *IEEE Trans. Intell. Transp. Syst.* **26**, 7465–7475 (2025).
27. Feng, S. et al. String stability for vehicular platoon control: definitions and analysis methods. *Annu. Rev. Control* **47**, 81–97 (2019).

28. Wang, J. et al. Parallel vision for long-tail regularization: initial results from IVFC autonomous driving testing. *IEEE Trans. Intell. Veh.* **7**, 286–299 (2022).
29. Engholm, A., Pernestål, A. & Kristoffersson, I. Cost analysis of driverless truck operations. *Transp. Res. Rec.* **2674**, 511–524 (2020).
30. *Truck driver salary in United States*, <https://www.indeed.com/career/truck-driver/salaries> (INDEED, 2025).
31. Wang, H., Hu, J. & Feng, Y. in *Proc. IEEE Intelligent Vehicles Symposium (IV)*. 2277-2282 (IEEE, 2024).
32. Wang, H. et al. Cut through traffic like a snake: cooperative adaptive cruise control with successive platoon lane-change capability. *J. Intell. Transp. Syst.* **28**, 1–22 (2022).
33. Chen, C. & Tomizuka, M. Modeling and control of articulated vehicles. *Appl. Sci.* **11**, 3663 (1997).

Acknowledgements

This paper is partially supported by National Key R&D Program of China (2022YFF0604905), National Natural Science Foundation of China (Grant No. 52302412 and 52372317), Yangtze River Delta Science and Technology Innovation Joint Force (No. 2023CSJGG0800), the Fundamental Research Funds for the Central Universities, Tongji Zhongte Chair Professor Foundation (No. 000000375-2018082), the Postdoctoral Fellowship Program (Grade B) of China Postdoctoral Science Foundation (GZB20230519), Shanghai Sailing Program (No. 23YF1449600), Shanghai Post-doctoral Excellence Program (No.2022571), China Postdoctoral Science Foundation (No.2022M722405), and the Science Fund of State Key Laboratory of Advanced Design and Manufacturing Technology for Vehicle (No. 32215011). We acknowledge OpenStreetMap for providing the open-access map data used in this study. The map data is available under the Open Database License. We acknowledge the use of the CARLA simulator to generate illustrative scenario examples for the figures. (Corresponding author: Haoran Wang).

Author contributions

J.H. and H.W. conceived the study and supervised the overall project. J.H. and Y.F. designed the research framework, developed the methodology. Y.F., M.L., and Y.Z. led the implementation of the methods, conducted experiments, and performed the primary data analysis. X.Z., Z.F., and J.L. assisted with data collection and supplementary analyses. All of the authors discussed the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s44172-025-00578-0.

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Peer review information *Communications Engineering* thanks Zhongbin Zhao, Marco Guerrieri and Pei Liu for their contribution to the peer review of this wo0072k. Primary Handling Editors: [Sasha Dong] and [Philip Coatsworth].